$\varepsilon(\omega,k)$, $\mu(\omega,k)$ and $\tilde{\varepsilon}(\omega,k)$ in isotropic medium.

Maxwell's eqns:

(1) $\vec{\nabla}\times\vec{E}+\frac{\partial}{\partial t}\vec{B}=0 \Rightarrow i\vec{k}\times\vec{E}(\omega,\vec{k})-i\omega\vec{B}(\omega,\vec{k})=0$

$\vec{k}\times\vec{E}=\omega\vec{B}$ for Fourier transforms of $\vec{E}(r,t)$ and $\vec{B}(r,t)$.

Similarly

$\vec{\nabla}\cdot\vec{B}=0 \Rightarrow \vec{k}\cdot\vec{B}=0$

$\vec{\nabla}\times\vec{B}=\mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t}+\mu_0\vec{J} \Rightarrow i\vec{k}\times\vec{B}=-\mu_0\varepsilon_0 i\omega\vec{E}+\mu_0(\vec{j}_{ext}+\vec{j}_{ind})$

$\vec{\nabla}\cdot\vec{E}=\frac{\rho}{\varepsilon_0}=\frac{\rho_{ext}+\rho_{ind}}{\varepsilon_0} \Rightarrow i\vec{k}\cdot\vec{E}=\frac{1}{\varepsilon_0}(\rho_{ext}+\rho_{ind})$

Using $\rho_{ind}=-\vec{\nabla}\cdot\vec{P} \Rightarrow \rho_{ind}(\omega,k)=-i\vec{k}\cdot\vec{P}(\omega,\vec{k})$

$\vec{j}_{ind}=\frac{\partial\vec{P}}{\partial t}+\vec{\nabla}\times\vec{M} \Rightarrow \vec{j}_{ind}=-i\omega\vec{P}+i\vec{k}\times\vec{M}$

We obtain in Fourier space (for linear medium)

$$\begin{cases} \vec{k}\times\vec{E}=\omega\vec{B} \\ \vec{k}\cdot\vec{B}=0 \\ i\vec{k}\times\vec{H}=i\frac{1}{\mu(\omega,k)}\vec{k}\times\vec{B}=-i\omega\vec{D}+\vec{j}_{ext} \\ \qquad =-i\omega\varepsilon\vec{E}+\vec{j}_{ext} \\ i\vec{k}\cdot\vec{D}=i\vec{k}\cdot\vec{E}\,\varepsilon=\rho_{ext} \end{cases}$$

System (I)

(3)

(4)

$\vec{M}=\chi_m\vec{H} \Rightarrow \vec{B}=\mu_0(1+\chi_m)\vec{H}=\mu\vec{H}$

While in terms of $\vec{\mathcal{D}}$:

$$\begin{cases} \vec{k} \times \vec{E} = \omega \vec{B} & \text{System (II)} \\ \vec{k} \cdot \vec{B} = 0 & \\ i\, \vec{k} \times \vec{B} = -i\omega\mu_0 \vec{\mathcal{D}} + \mu_0 \vec{j}_{ext} & (3) \\ i\, \vec{k} \cdot \vec{\mathcal{D}} = i\, \vec{k} \cdot \vec{E}\, \tilde{\varepsilon} = \rho_{ext} & (4) \end{cases}$$

To connect two descriptions we need to understand structure of $\tilde{\varepsilon}(\omega, \vec{k})$: can form tensor even in isotropic medium

generally $\tilde{\varepsilon}_{\alpha\beta}(\omega, \vec{k}) = A_1(\omega, \vec{k})\, \delta_{\alpha\beta} + A_2(\omega, \vec{k})\, \dfrac{k_\alpha k_\beta}{k^2}$

where $A_{1,2}(\omega, \vec{k}) = A_{1,2}(\omega, k)$

$\hookrightarrow k = |\vec{k}|$

$$\mathcal{D}_\alpha = \tilde{\varepsilon}_{\alpha\beta} E_\beta \Rightarrow \mathcal{D}_\alpha = A_1 E_\alpha + A_2 \frac{k_\alpha (\vec{k} \cdot \vec{E})}{k^2}$$

$$\vec{\mathcal{D}} = A_1 \vec{E} + A_2 \frac{\vec{k}(\vec{k} \cdot \vec{E})}{k^2}$$

That is, response for <u>transverse</u> and <u>longitudinal</u> fields are different!

For $\vec{k} \cdot \vec{E} = 0$, so that $\vec{E} \perp \vec{k}$, (transverse)

$$\vec{\mathcal{D}} = A_1 \vec{E} = \tilde{\varepsilon}^{tr}(\omega, \vec{k})\, \vec{E}$$

For $\vec{E} \parallel \vec{k}$, $\vec{\mathcal{D}} = (A_1 + A_2)\, \vec{E} = \tilde{\varepsilon}^{l}(\omega, k)\, \vec{E}$

longitudinal dielectric function

$$A_2 = \tilde{\varepsilon}^{\ell} - A_1 = \tilde{\varepsilon}^{\ell} - \tilde{\varepsilon}^{tr}$$

$$\Rightarrow \underset{(\omega,k)}{\tilde{\varepsilon}} = \tilde{\varepsilon}^{tr}\,\delta_{\alpha\beta} + \left(\tilde{\varepsilon}^{\ell} - \tilde{\varepsilon}^{tr}\right)\frac{k_\alpha k_\beta}{k^2}$$

$$= \underset{(\omega,k)}{\tilde{\varepsilon}^{tr}}\left(\delta_{\alpha\beta} - \frac{k_\alpha k_\beta}{k^2}\right) + \underset{(\omega,k)}{\tilde{\varepsilon}^{\ell}}\,\frac{k_\alpha k_\beta}{k^2}$$

Compare eqns (4) in systems I and II:

$$i\,\vec{k}\cdot\left(\vec{E}\,\varepsilon\right) = \rho_{ext}$$

$$i\,k_a D_a = \rho_{ext}$$

$D_a = \varepsilon(\omega,k)\,E_a$ in isotropic medium.

and $i\,\vec{k}\cdot\left(\vec{E}\,\tilde{\varepsilon}\right) = \rho_{ext}$

$$i\,\vec{k}\cdot\vec{\mathcal{D}} = \rho_{ext}$$

In components: $i\,k_a \mathcal{D}_a = i\,k_a\left(\tilde{\varepsilon}_{ab}\,E_b\right) =$

$$= i\,k_a\left\{\left(\delta_{ab} - \frac{k_a k_b}{k^2}\right)\tilde{\varepsilon}^{tr} + \tilde{\varepsilon}_{\ell}\,\frac{k_a k_b}{k^2}\right\}E_b = i\,k_b E_b\,\tilde{\varepsilon}_{\ell} = \rho_{ext}$$

but $k_a\left(\delta_{ab} - \frac{k_a k_b}{k^2}\right) = k_b - \frac{k_a k_a k_b}{k^2} =$

$$= k_b - k_b = 0$$

while $k_a\,\frac{k_a k_b}{k^2} = \frac{k_a k_a}{k^2}\,k_b = k_b$

Therefore $\varepsilon(\omega,k) = \tilde{\varepsilon}_\ell(\omega,k)$ (longitudinal)

Now look at eq (3):

I $\quad \frac{i}{\mu}\vec{k}\times\vec{B} = -i\omega\,\varepsilon\,\vec{E} + \vec{j}_{ext} \Rightarrow i\,\frac{\mu_0}{\mu}\,\vec{k}\times\vec{B} = -i\mu_0\omega\varepsilon\vec{E} + \mu_0\vec{j}_{ext}$

II $\quad i\vec{k}\times\vec{B} = -i\omega\,\underbrace{\mu_0\,\cancel{\varepsilon\vec{E}}}_{\vec{\mathcal{D}}} + \mu_0\vec{j}_{ext}$

$$\vec{\mathcal{D}} = \tilde{\varepsilon}_{tr}\left(\vec{E} - \frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}\right) + \tilde{\varepsilon}_\ell\,\frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}$$

Subtract(μ_0 I) from II:

$$i\left(1-\frac{\mu_0}{\mu}\right)\vec{k}\times\vec{B} = -i\omega\mu_0\left[\tilde{\varepsilon}_{tr}\left(\vec{E}-\frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}\right) + \tilde{\varepsilon}_\ell\,\frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2} - \varepsilon\vec{E}\right]$$

but $\varepsilon = \tilde{\varepsilon}_\ell \Rightarrow$

$$i\left(1-\frac{\mu_0}{\mu}\right)\vec{k}\times\vec{B} = -i\omega\mu_0\left(\tilde{\varepsilon}_{tr}-\tilde{\varepsilon}_\ell\right)\left(\vec{E}-\frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}\right)$$

From Faraday's law: $\vec{B} = \frac{1}{\omega}\vec{k}\times\vec{E} \Rightarrow \vec{k}\times\vec{B} = \frac{1}{\omega}\vec{k}\times(\vec{k}\times\vec{E})$

$$= \frac{1}{\omega}\left\{\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E}\right\} =$$

$$= -\frac{k^2}{\omega}\left(\vec{E} - \frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}\right)$$

Therefore

$$\cancel{i}\left(1-\frac{\mu_0}{\mu}\right)\frac{k^2}{\omega} = \cancel{i}\omega\mu_0\left(\tilde{\varepsilon}_{tr}-\tilde{\varepsilon}_l\right)$$

$$1-\frac{\mu_0}{\mu(k,\omega)} = \frac{\omega^2}{k^2}\mu_0\left(\tilde{\varepsilon}_{tr}-\tilde{\varepsilon}_l\right) = \frac{\omega^2}{c^2k^2}\left(\frac{\tilde{\varepsilon}_{tr}-\tilde{\varepsilon}_l}{\varepsilon_0}\right)$$

$$\frac{\mu(k,\omega)}{\mu_0} = \left\{1-\frac{\omega^2}{c^2k^2}\left[\frac{\tilde{\varepsilon}_{tr}(k,\omega)-\tilde{\varepsilon}_l(k,\omega)}{\varepsilon_0}\right]\right\}^{-1}$$

We observe that difference between transverse $\tilde{\varepsilon}_{tr}(k,\omega)$ and longitudinal $\tilde{\varepsilon}_l(k,\omega)$ dielectric functions is due to magnetic permiability $\mu(k,\omega)=\mu_0\left(1+\chi_m(k,\omega)\right)$.

If $\mu(k,\omega)=\mu_0$, then there is no difference, $\tilde{\varepsilon}_{tr}=\tilde{\varepsilon}_l=\varepsilon(k,\omega)$.

Therefore ~~any person~~ every material is characterized by two material-dependent quantities:

1) at $\omega=0$ (static limit) we need $\varepsilon=\varepsilon_0(1+\chi_e)$
and $\mu=\mu_0(1+\chi_m)$

2) at finite $\omega\neq 0$, we need $\varepsilon(k,\omega)$ and $\mu(k,\omega)$

OR

we can proceed without $\vec{B}$ at all, since $\omega\vec{B}=\vec{k}\times\vec{E}$. Then we need $\tilde{\varepsilon}_{tr}(k,\omega)$ and $\tilde{\varepsilon}_l(k,\omega)$.

Can also express $\tilde{\varepsilon}$ in terms of ε, μ:

$$\tilde{\varepsilon}_{\alpha\beta} = \left(\delta_{\alpha\beta} - \frac{k_\alpha k_\beta}{k^2}\right)\tilde{\varepsilon}_{tr} + \frac{k_\alpha k_\beta}{k^2}\underset{\varepsilon}{\underset{\shortparallel}{\tilde{\varepsilon}_\ell}}$$

$$\frac{\tilde{\varepsilon}_{tr}}{\varepsilon_0} = \frac{c^2 k^2}{\omega^2}\left(1 - \frac{\mu_0}{\mu}\right) + \frac{\tilde{\varepsilon}_\ell}{\varepsilon_0}$$

$$\Rightarrow \tilde{\varepsilon}_{tr} = \frac{c^2 k^2}{\omega^2}\varepsilon_0\left(1 - \frac{\mu_0}{\mu}\right) + \tilde{\varepsilon} = \frac{k^2}{\omega^2}\left(\frac{1}{\mu_0} - \frac{1}{\mu}\right) + \tilde{\varepsilon}$$

$$c^2\varepsilon_0 = \frac{\varepsilon_0}{\varepsilon_0\mu_0} = \frac{1}{\mu_0}$$

$$\Rightarrow \tilde{\varepsilon}_{\alpha\beta} = \delta_{\alpha\beta}\,\varepsilon + \frac{k^2}{\omega^2}\left(\frac{1}{\mu_0} - \frac{1}{\mu}\right)\left(\delta_{\alpha\beta} - \frac{k_\alpha k_\beta}{k^2}\right).$$

scalar functions of $(\omega, \vec{k})$

Oscillations of EM field in the medium.

These are oscillations that take place in the absence of external sources $\Rightarrow \rho_{ext} = 0, \ \vec{j}_{ext} = 0$

$\Rightarrow$ (Gauss' law) $\vec{k} \cdot \vec{D} = 0$

$$\Rightarrow \quad A_1 \vec{k} \cdot \vec{E} + A_2 \frac{\vec{k} \cdot \vec{k} (\vec{k} \cdot \vec{E})}{k^2} = 0$$

$$\underbrace{(A_1 + A_2)}_{\tilde{\varepsilon}^{\ell}(\omega, k)} (\vec{k} \cdot \vec{E}) = 0$$

$$\Rightarrow \quad \boxed{\tilde{\varepsilon}^{\ell}(\omega, \vec{k}) = 0}$$

longitudinal oscillations.

Only electric field is involved.

Simplest model (Drude) of non-magnetic medium ($\mu(\omega, k) = \mu_0$)

$\tilde{\varepsilon}_{\alpha\beta}(\omega, \vec{k}) = \delta_{\alpha\beta} \ \varepsilon(\omega, \vec{k})$ so that there is no difference between longitudinal ($\tilde{\varepsilon}^{\ell}$) and transverse ($\tilde{\varepsilon}^{tr}$) components of $\tilde{\varepsilon}$.

Then $\varepsilon(\omega, \vec{k}) \cong \varepsilon_0 \left(1 + \frac{i \sigma(\omega)}{\varepsilon_0 \omega}\right)$ for conductors

$$\sigma(\omega) = \frac{\sigma_0}{1 - i\omega\tau} \xrightarrow[\omega\tau \gg 1]{} \frac{i \sigma_0}{\omega \tau} \quad \text{(high-frequency limit)}$$

so that $\varepsilon(\omega) = \left(1 + \frac{i}{\varepsilon_0 \omega} \frac{i \sigma_0}{\omega \tau}\right) \varepsilon_0 = \varepsilon_0 \left(1 - \frac{\sigma_0}{\varepsilon_0 \omega^2 \tau}\right)$

Therefore we obtain $\varepsilon_0 \left(1 - \frac{\sigma_0}{\varepsilon_0 \omega^2 \tau}\right) = 0$

$$\omega^2 = \frac{\sigma_0}{\varepsilon_0 \tau} = \omega_p^2 = \frac{e^2 n}{\varepsilon_0 m} \quad \text{Plasma frequency.}$$

$$\sigma_0 = \varepsilon_0 \omega_p^2 \tau$$

$$\vec{k} \times \vec{E} = \omega \vec{B}$$

$$i\,\vec{k} \times \vec{B} = -i\omega\mu_0 \vec{\mathcal{D}} + \mu_0 \underbrace{\vec{j}_{ext}}_{0}$$

$$\Rightarrow \vec{k} \times \vec{B} = -\omega\mu_0 \vec{\mathcal{D}}$$

but $\vec{k} \times \vec{B} = \frac{1}{\omega}\, \vec{k} \times (\vec{k} \times \vec{E}) = \left(\vec{k}(\vec{k}\cdot\vec{E}) - k^2 \vec{E}\right)\frac{1}{\omega}$

$$\vec{\mathcal{D}} = \underbrace{A_1}_{\tilde{\varepsilon}^{tr}} \vec{E} + \underbrace{A_2}_{\tilde{\varepsilon}^{l} - \tilde{\varepsilon}^{tr}} \frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2} = \tilde{\varepsilon}^{tr}\left(\vec{E} - \frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}\right) + \tilde{\varepsilon}^{l}\frac{\vec{k}(\vec{k}\cdot\vec{E})}{k^2}$$

$$\Rightarrow \frac{1}{\omega}\left(\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E}\right) = -\omega\mu_0\left\{\frac{\tilde{\varepsilon}^{tr}}{k^2}\left(k^2\vec{E} - \vec{k}(\vec{k}\cdot\vec{E})\right) + \frac{\tilde{\varepsilon}^{l}}{k^2}\vec{k}(\vec{k}\cdot\vec{E})\right\}$$

$$\Rightarrow \left(\frac{1}{\omega} - \frac{\omega\mu_0}{k^2}\tilde{\varepsilon}^{tr}\right)\left(\vec{k}(\vec{k}\cdot\vec{E}) - k^2\vec{E}\right) = -\mu_0\frac{\omega}{k^2}\tilde{\varepsilon}^{l}\,\vec{k}(\vec{k}\cdot\vec{E})$$

$\Rightarrow$ For transverse electric field $\vec{E}^{\perp}$, such that $\vec{k}\cdot\vec{E}^{\perp} = 0$,

we obtain $\omega^2\mu_0\tilde{\varepsilon}^{tr}(\omega,\vec{k}) = k^2$

If we separate ε_0 from $\tilde{\varepsilon}^{tr}(\omega,\vec{k}) = \varepsilon_0\,\varepsilon^{\perp}(\omega,\vec{k})$

$$\Rightarrow \omega^2\varepsilon^{\perp}(\omega,k) = \frac{1}{\mu_0\varepsilon_0}k^2 = c^2k^2$$

These are transverse oscillations!

For non-magnetic media, $\tilde{\varepsilon}^{tr} = \varepsilon \Rightarrow \tilde{\varepsilon}^{tr} = \varepsilon_0(1+\chi_e) \Rightarrow$ obtain $\boxed{1+\chi_e(\omega,k) = \frac{c^2k^2}{\omega^2}}$

* dispersion of EM waves in dielectric